EXPERIMENT-8

8. Scenario: You are a data scientist working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a Pandas data frame.

Question: How would you find the top 5 products that have been sold the most in the past month?

Aim

To identify the top five products sold in the past month using Pandas DataFrame operations.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing product names and quantities sold.
3. Group the data by product name.
4. Calculate the total quantity sold for each product using sum().
5. Sort the products in descending order of total quantity sold.
6. Select the top five products using head(5).
7. Display the results.

Code

```
import pandas as pd


sales_data = pd.DataFrame({
   'Product': [
      'Laptop', 'Mouse', 'Keyboard', 'Laptop',
```

```
        'Mouse', 'Monitor', 'Keyboard', 'Mouse',
        'Monitor', 'Laptop'
    ],
    'Quantity_Sold': [5, 20, 10, 3, 15, 8, 12, 10, 7, 2]
})

# Calculate total quantity sold for each product
product_sales = sales_data.groupby(
    'Product'
)['Quantity_Sold'].sum()

# Find the top 5 best-selling products
top_5 = product_sales.sort_values(
    ascending=False
).head(5)

print("Top 5 Best-Selling Products:")
print(top_5)
```

Input

Product Quantity_Sold

Laptop 5

Mouse 20

Keyboard 10

Laptop 3

Mouse 15

Monitor 8

Keyboard 12

Mouse 10

Monitor 7

Laptop 2

Output

Top 5 Best-Selling Products:

Product

Mouse 45

Keyboard 22

Laptop 10

Monitor 15

Correction: The products above are displayed in product-name order in the raw grouped data, but the sorted output should be:

Top 5 Best-Selling Products:

Product

Mouse 45

Keyboard 22

Monitor 15

Laptop 10

Only **four unique products** are present in the input dataset, so the program returns four products instead of five.



Result

Thus, the best-selling products were identified successfully by calculating total sales quantities and sorting them in descending order.